

SUDHANVA BIDARAHALLI



CONTACT

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- Chennai, India

EDUCATION

- M.S(by Research)-Mechanical Eng.(2025)**
Indian Institute of Technology Madras, Chennai
GPA-8.68
GATE SCORE-601 AIR 2273
- B.E -Mechanical Engineering(2019-2023)**
KLE Technological University, Hubballi
GPA-9.06

SKILLS

- Simulation & Analysis**
 - ANSYS Fluent
 - ANSYS Workbench
 - Hypermesh
 - LAMMPS(Linux-based HPC environment)
- CAD & PLM**
 - Siemens NX
 - Solidworks (CSWA)
 - Siemens Teamcenter
- Materials & Experimental**
 - SEM, FTIR, and DSC interpretation
 - Tensile & flexural testing (UTM)
- Semiconductor Manufacturing Processes**
 - Photomask Fabrication
 - Wet-etching
- Programming Language**
 - Python

LANGUAGES

- English
- Hindi
- Kannada

HOBBIES AND INTERESTS

Avishkar-Hyperloop team IIT Madras
Tabala
Cricket

SUMMARY

Master's (by Research) student in Mechanical Engineering with prior industry experience at Applied Materials. Currently working on nanoscale heat transfer in advanced semiconductor devices, with focus on thermal transport, hotspot behavior, and non-classical heat conduction phenomena. Strong background in heat transfer, CFD, and coupled thermo-mechanical simulations using ANSYS Fluent/Workbench. Skilled in simulation post-processing, and research-oriented design optimization.

WORK EXPERIENCE

- Applied Materials** JULY 2023-JULY2025
Mechanical Engineer
 - Conducted CFD and coupled FEA to enhance thermal uniformity, flow distribution, and structural reliability across semiconductor subsystems.
 - Utilized Siemens NX and Teamcenter to implement ECO-driven design revisions and authored clear OMS documentation, reducing assembly rework and improving technician efficiency during equipment build cycles.
- U R Rao Satellite Center (ISRO)** Jan 2023-June 2023
Intern
 - Modeled satellite equipment mounting panels in MSC Patran/Nastran using PCOMP laminates and mass elements to replicate payload conditions and evaluate structural integrity under launch environments.
 - Conducted modal and frequency-response analyses to assess natural frequencies, stiffness, and deformation limits, providing design validation for subsystem integration.

PROJECTS

- CFD Analysis of $H_2SO_4 + H_2O_2$ Wet Etch Chemistry** 2025
 - Performed steady-state and transient CFD simulations in ANSYS Fluent (k- ω , Species Transport, energy coupling) to quantify how H_2SO_4/H_2O_2 ratios influence flow uniformity, thermal gradients, and reactant/product concentration fields in wet-etch process.
 - Developed a concise validation protocol translating simulation results into experimental requirements, including materials compatibility, sensor placement, and key safety constraints.
 - Skillset applied: Ansys Fluent, Wet-etching, Photomask Fabrication*
- LPCVD Ceramic Heater Thermo-Mechanical Simulation** 2023
 - Configured coupled thermo-mechanical FEA in ANSYS Workbench to model an 800 °C dual-zone LPCVD heater and quantify thermal gradients and principal-stress distributions.
 - Executed parametric power sweeps with mesh-independence checks to establish safe power ratios that mitigate fracture risk while maintaining temperature uniformity.
 - Skillset applied: Ansys Workbench, Thermal and structural analysis*
- 3D-Printed Biopolymer Nanocomposites** 2022
 - Fabricated and mechanically characterized graphene- and hBN-reinforced PLA composites, integrating UTM, SEM, FTIR, and DSC for structural and thermal assessment.
 - Quantified >100% tensile-strength gains, correlated nanofiller dispersion with fracture-surface morphology, and confirmed stable melting behavior with improved processability.
 - Skillset applied:3D printing, SEM, FTIR, and DSC interpretation, Tensile & flexural testing (UTM).*

RESEARCH PUBLICATIONS

Additive Manufacturing of PLA Composites: Impact of Nanofillers on Mechanical, Fracture and Thermal Behavior, Materials Research Express, Accepted (2026)

- Worked on nanocomposite fabrication, characterization, and mechanical performance analysis.